ASSIGNMENT 01

Footprinting & Reconnaissance

Practical lab report: recon-ng, Google dorks, subdomain enumeration, Maltego and theHarvester

SUBMISSION DETAILS

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Course	Cyber Policy
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Submission date	01/11/2026
Environment	Kali Linux 2026.2 (VMware virtual machine) & Win 11

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1. Overview

This report covers five footprinting and reconnaissance tasks, all run on a Kali Linux 2026.2 virtual machine (VMware) using passive methods only. Targets were limited to the domains named in the assignment (tesla.com and microsoft.com), and every API key was created from the author's own free account.

Task	Topic	Main tool	Evidence
1	Information gathering on tesla.com	recon-ng v5.1.2	Screen recording
2	Google dorks	Google search operators	Screenshots
3	Microsoft subdomains	curl, host, dig, nslookup, whois	Screen recording
4	Maltego installation	Maltego 4.13.0	Screenshots
5	theHarvester with API keys	theHarvester 4.10.1	Screen recording

Submission folder layout

```
Assignment01-Maryam Batool/
  Assignment01_Report.pdf
  Task1/  recording + tesla_report.html + tesla_report.txt
  Task2/  screenshots of each dork
  Task3/  recording + task3_results.txt + ms_subs.txt
  Task4/  screenshots
  Task5/  recording + task5_output.json + task5_output.xml + task5_results.txt
```

2. Task 1: recon-ng on tesla.com

recon-ng v5.1.2 was used to collect subdomains and hosts for tesla.com in a dedicated workspace. The results were exported as tesla_report.html and converted to tesla_report.txt.

Commands

```
recon-ng
workspaces create tesla
marketplace install recon/domains-hosts/hackertarget
marketplace install recon/domains-hosts/certificate_transparency
marketplace install recon/hosts-hosts/resolve
marketplace install reporting/html
db insert domains            # domain: tesla.com
modules load recon/domains-hosts/hackertarget
run
modules load recon/domains-hosts/certificate_transparency
run
modules load recon/hosts-hosts/resolve
run
show hosts
modules load reporting/html
options set CREATOR <name>
options set CUSTOMER tesla.com
options set FILENAME /home/kali/tesla_report.html
run
```

Steps explained

1. Created the workspace "tesla" so results stay separate from other targets.
2. Installed four marketplace modules: two discovery modules, one DNS resolver and one reporting module.
3. Added tesla.com to the domains table, then ran hackertarget and certificate_transparency to find subdomains.
4. Ran resolve to map the discovered hosts to IP addresses and listed them with show hosts.
5. Generated the HTML report with reporting/html and converted it to text with html2text.

Output files: tesla_report.html, tesla_report.txt

Evidence

Screen recording	Task1/ Screen Recording Task1

3. Task 2: Google dorks

Ten search-operator queries were run on Google, with one screenshot per query. Each operator narrows the results by domain type, file type, page title or URL.

#	Goal	Query
1	Educational websites	`site:edu`
2	Government websites	`site:gov`
3	PDF documents	`filetype:pdf cybersecurity`
4	Word documents	`filetype:docx cybersecurity`
5	"cybersecurity" in the title	`intitle:cybersecurity`
6	"about" in the URL	`inurl:about`
7	Exact phrase	`"network security fundamentals"`
8	PDFs on educational sites	`site:edu filetype:pdf`
9	Admission-related pages	`site:edu inurl:admission`
10	Cybersecurity on government sites	`site:gov cybersecurity`

Observation: a bare "filetype:pdf" returned "did not match any documents", because Google needs at least one search keyword alongside the filetype: operator. Adding a keyword such as "cybersecurity" fixed it.

Evidence

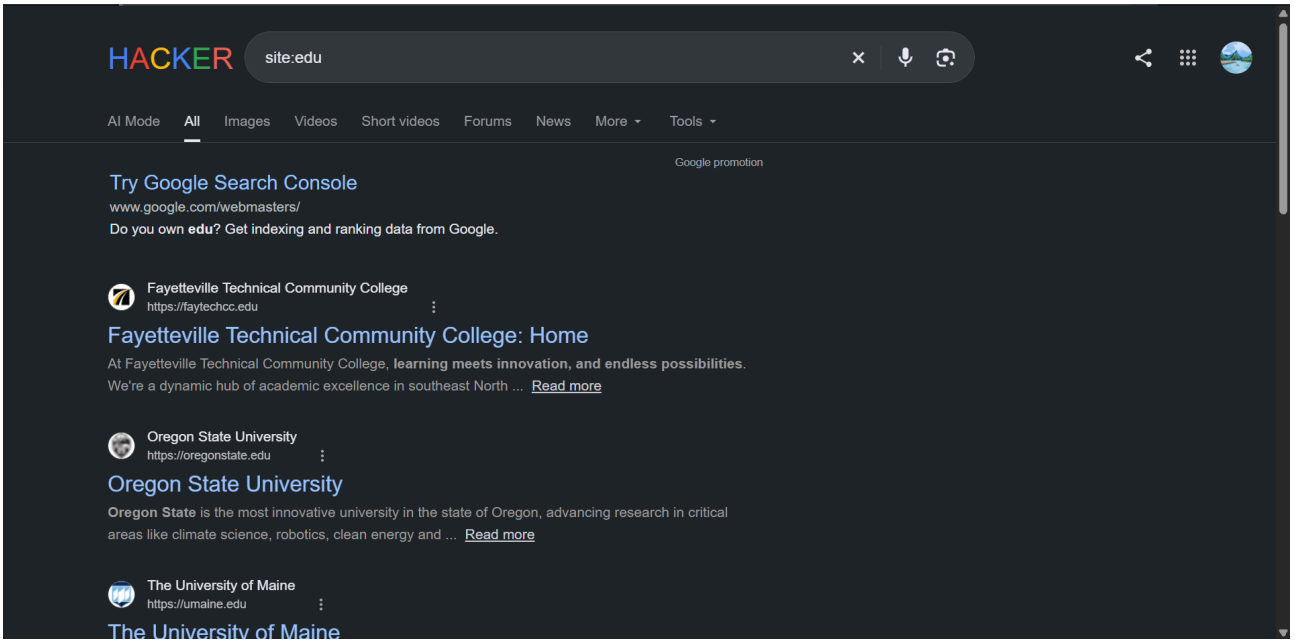


Figure 3.1: site:edu

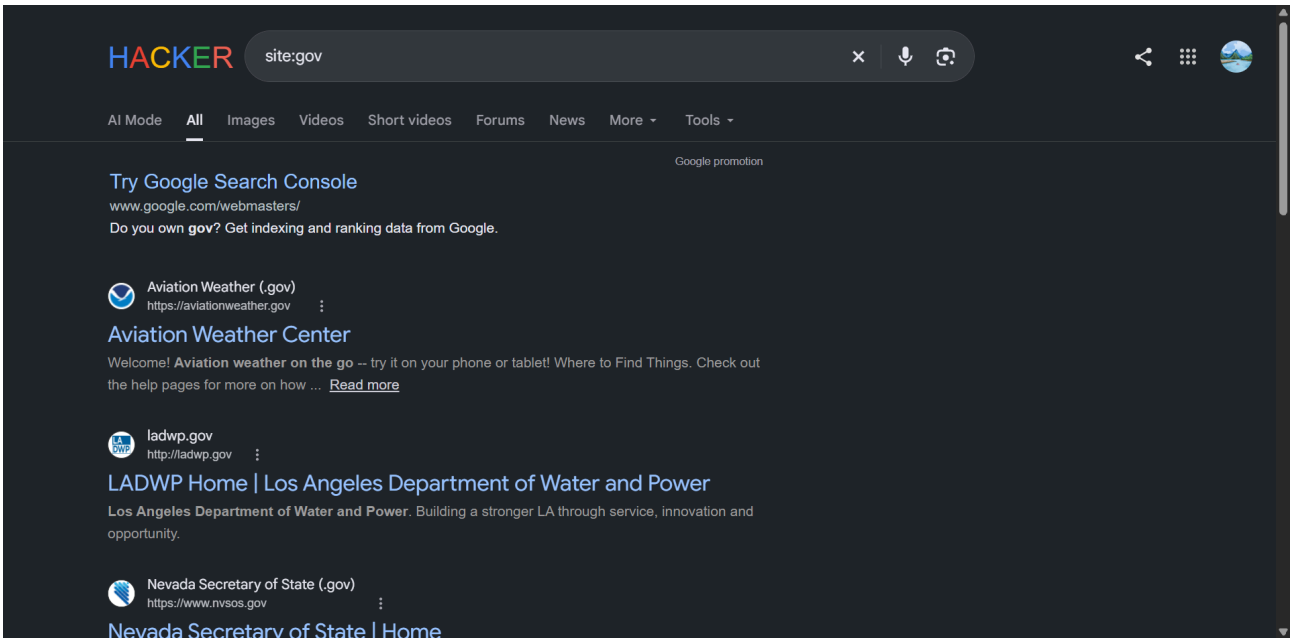


Figure 3.2: site:gov

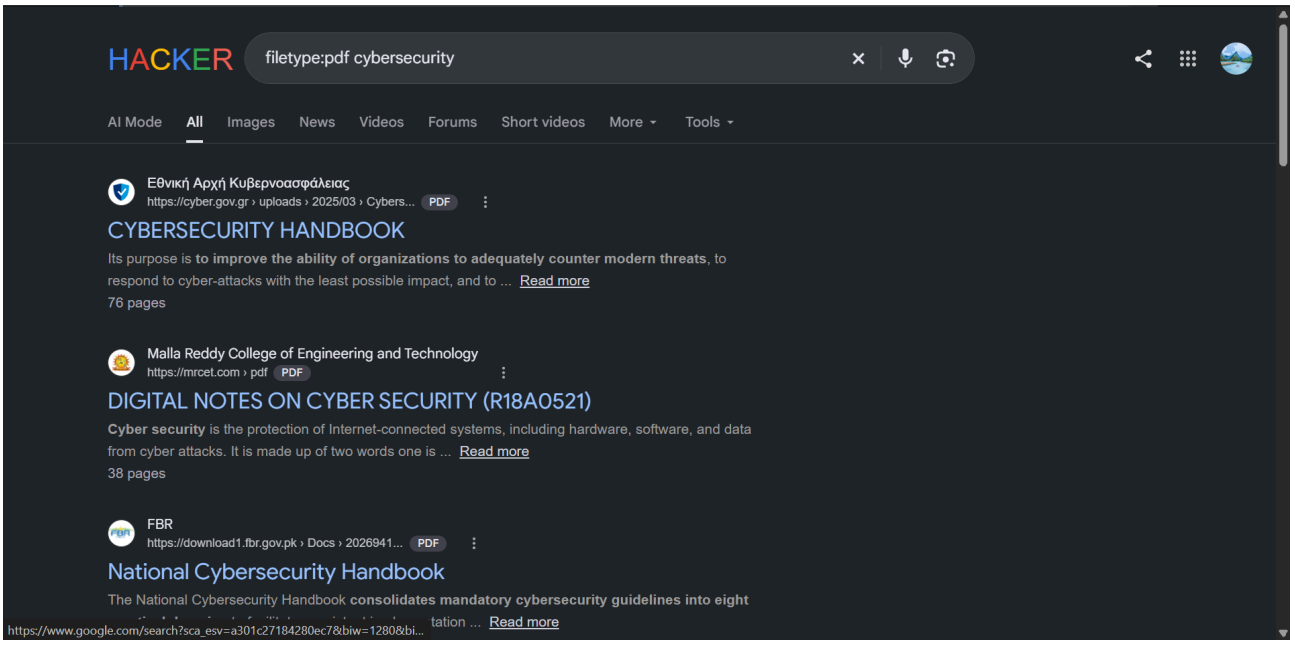


Figure 3.3: filetype:pdf cybersecurity

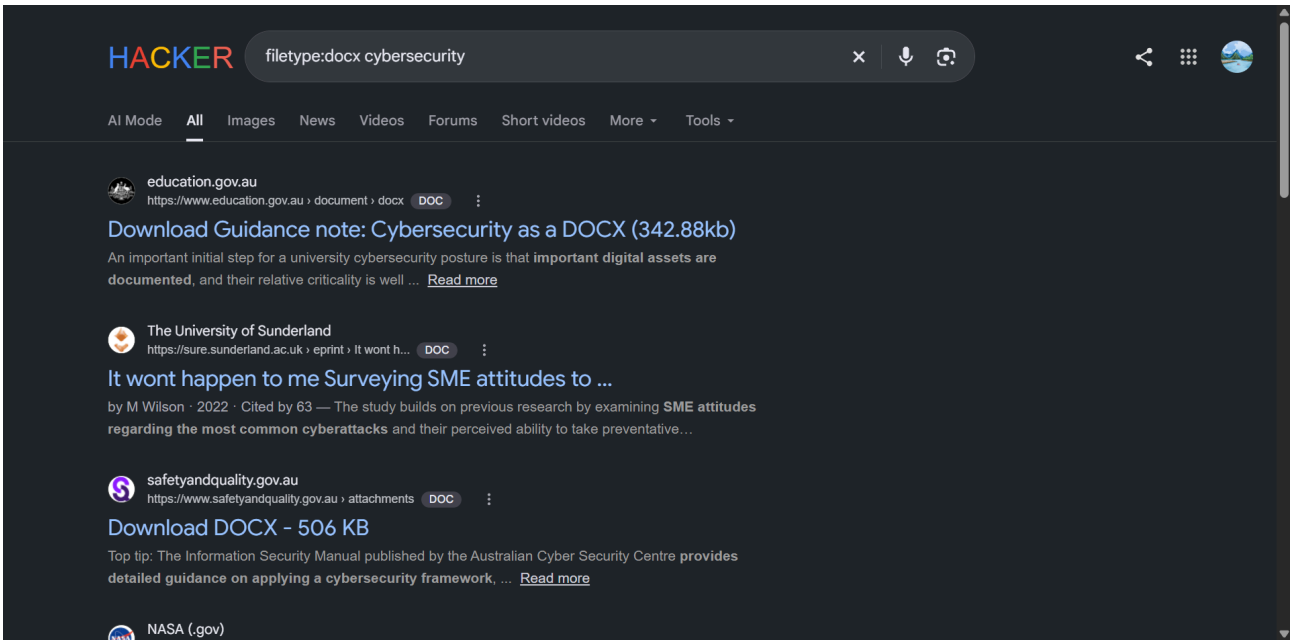


Figure 3.4: filetype:docx cybersecurity

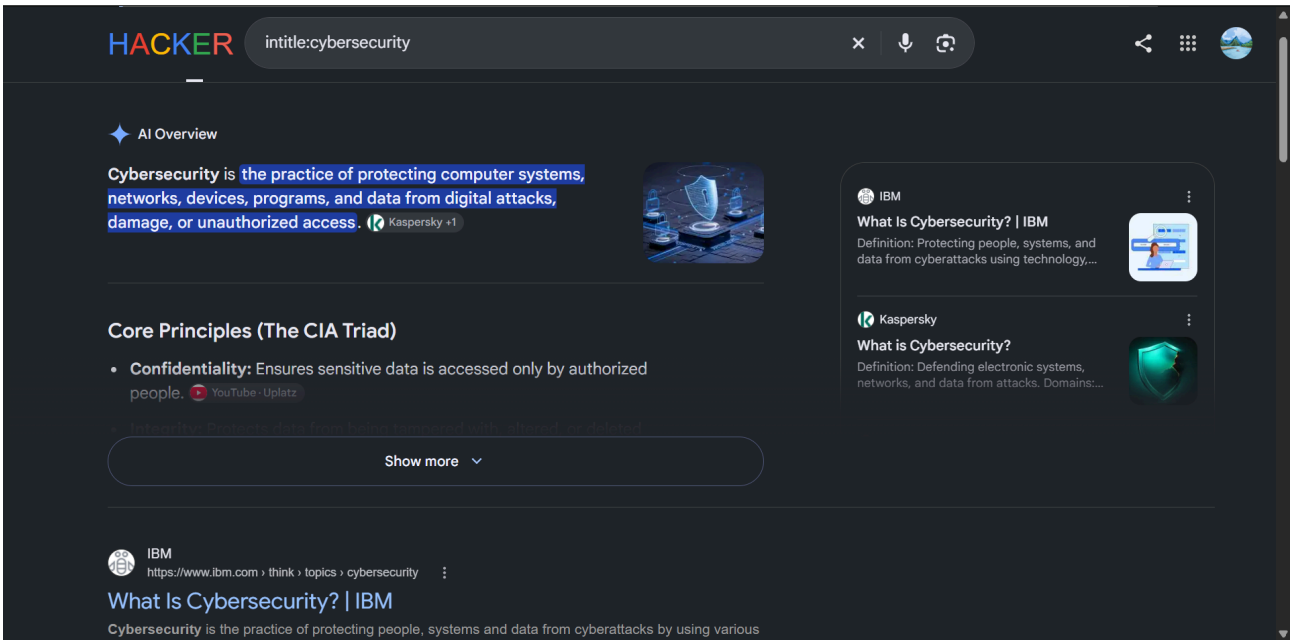


Figure 3.5: intitle:cybersecurity

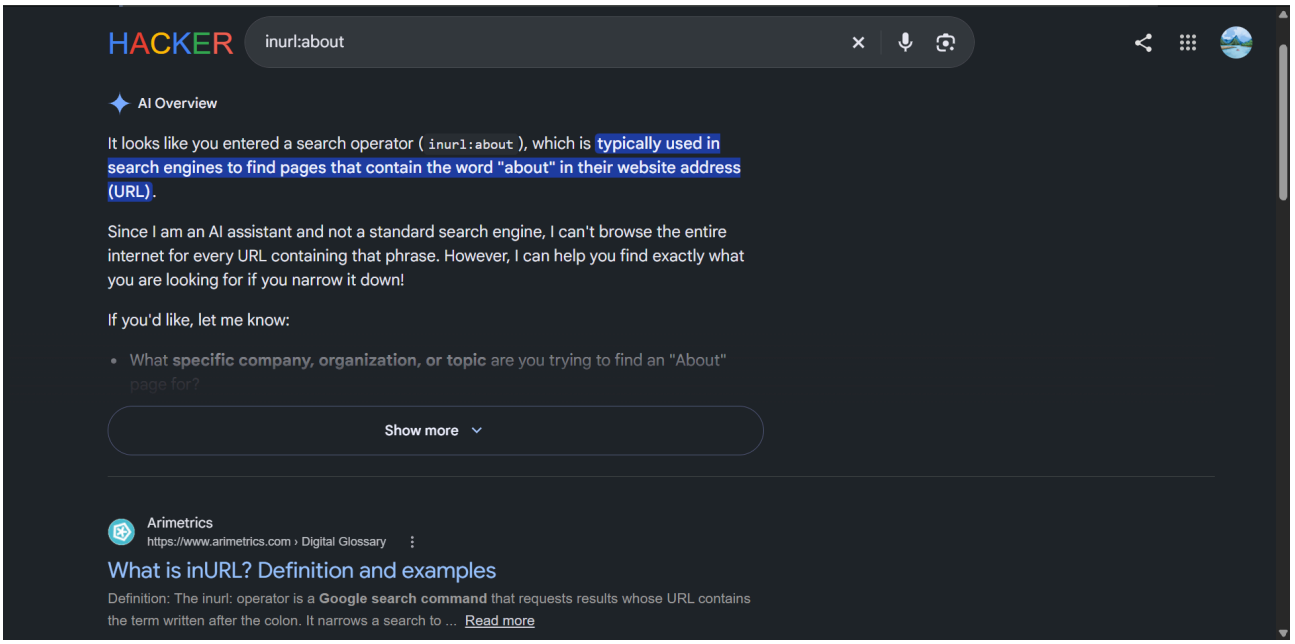


Figure 3.6: inurl:about

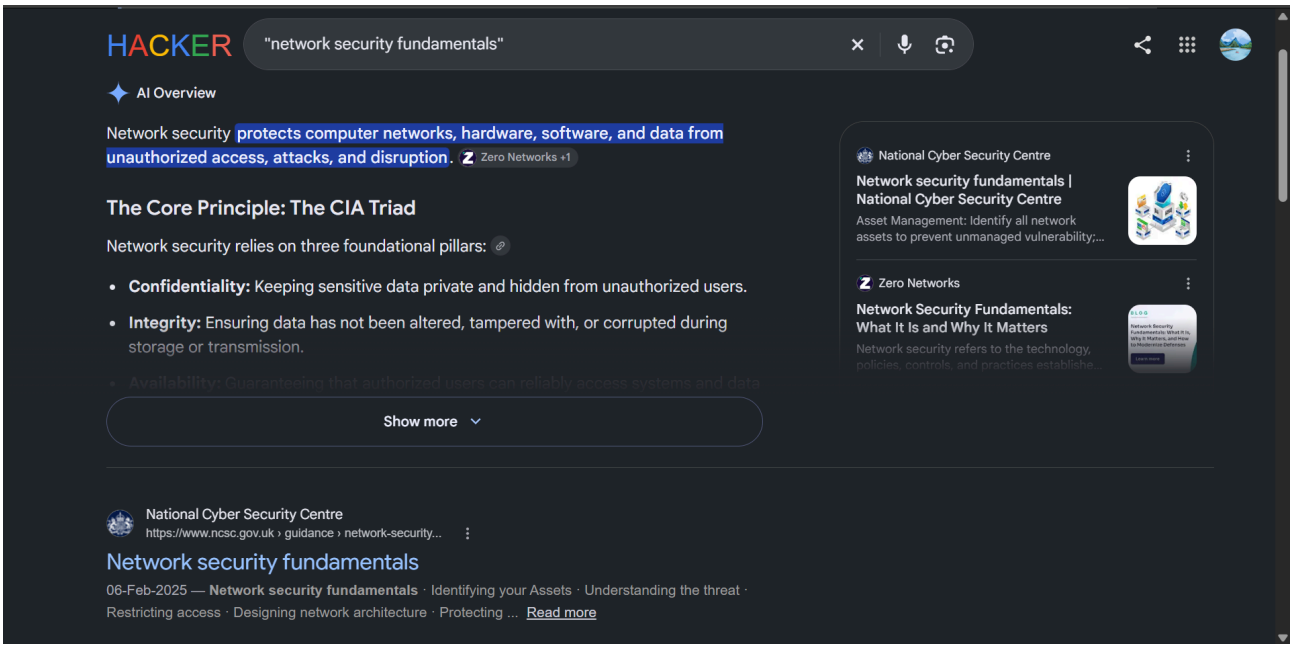


Figure 3.7: "network security fundamentals"

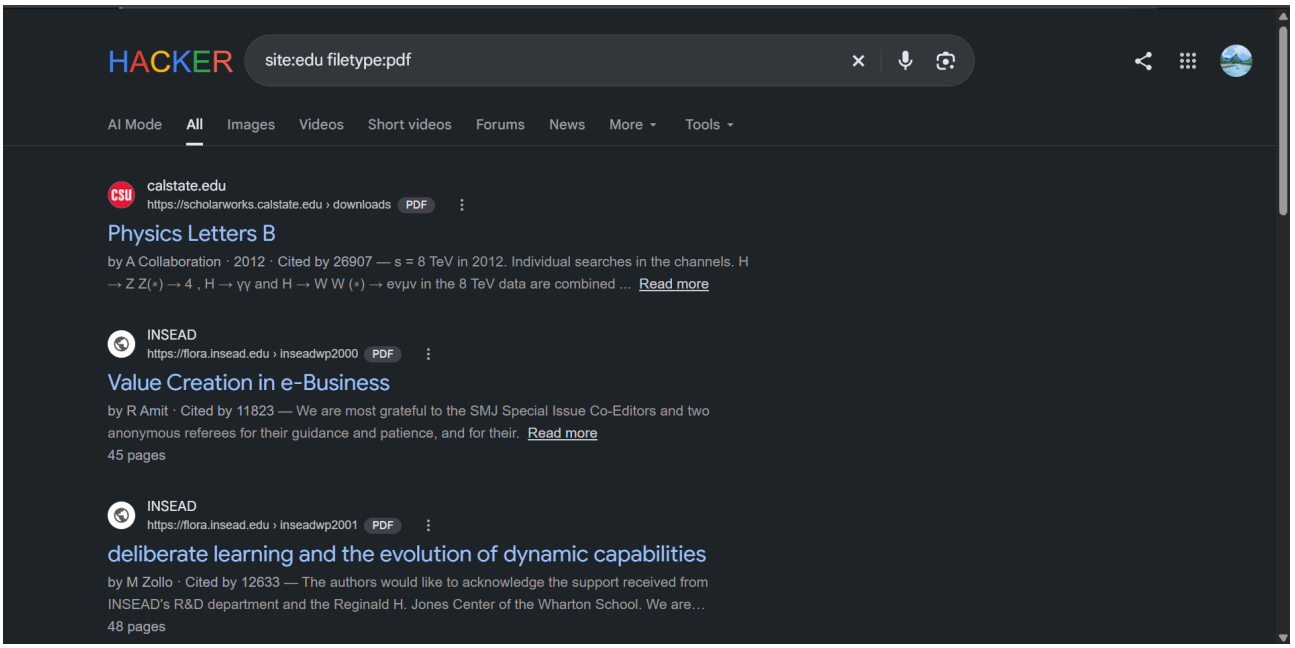


Figure 3.8: site:edu filetype:pdf

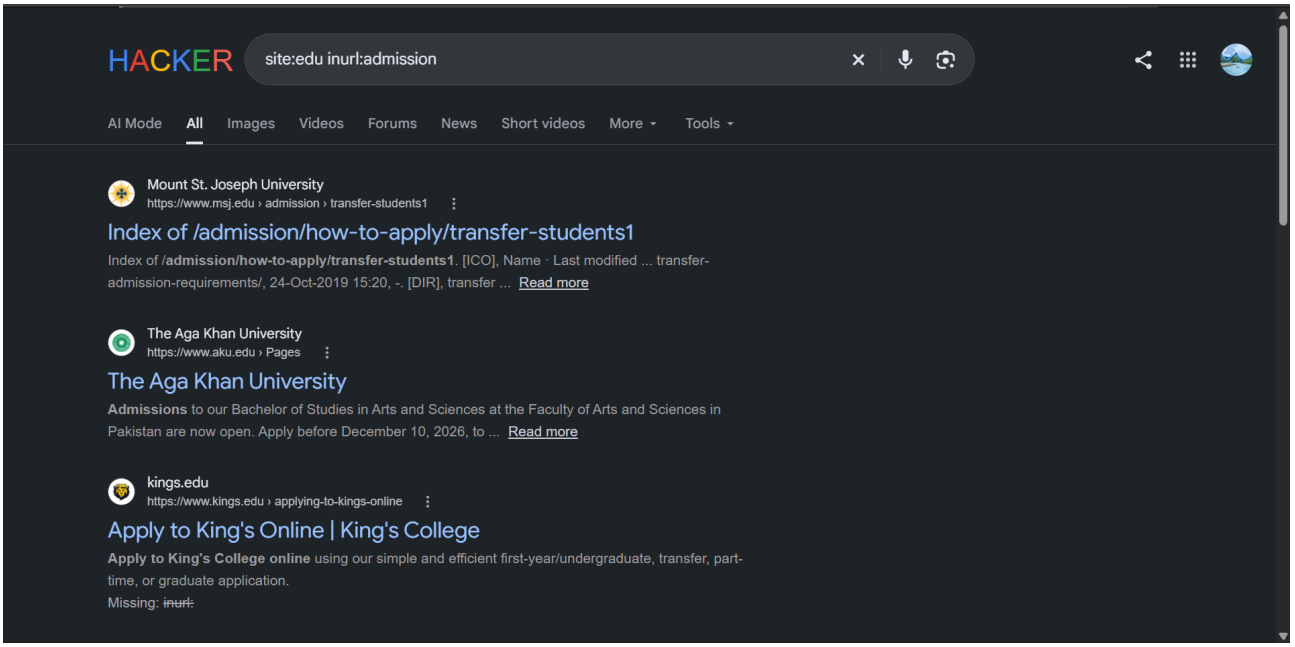


Figure 3.9: site:edu inurl:admission

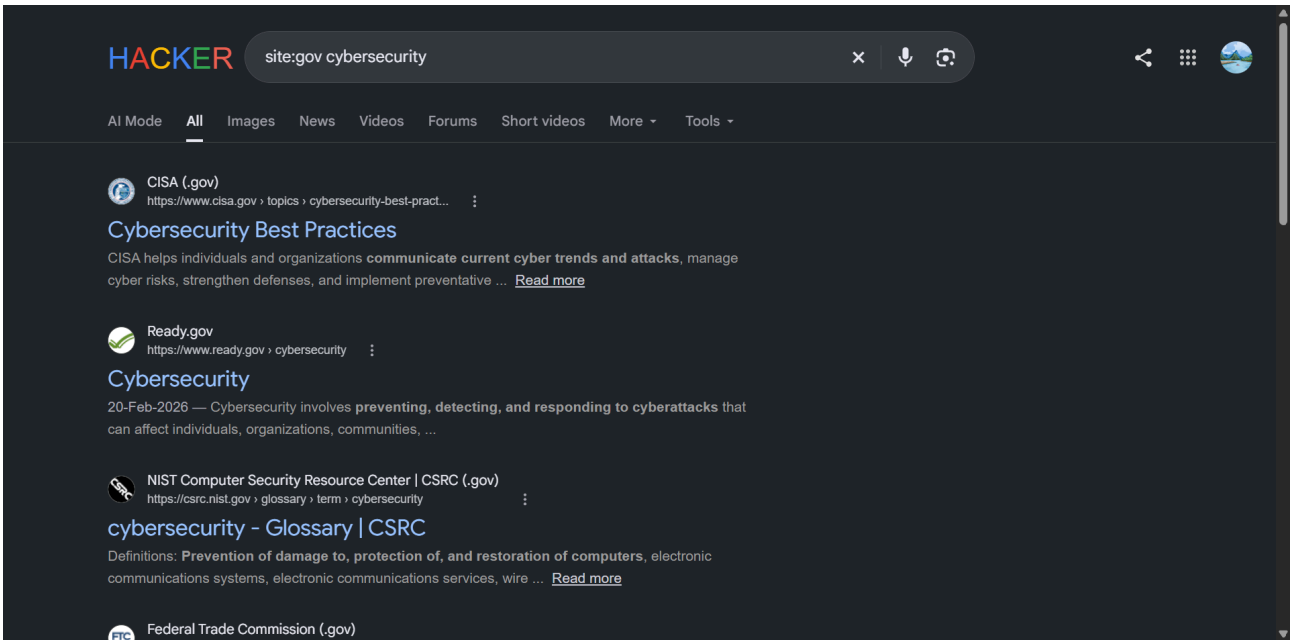


Figure 3.10: site:gov cybersecurity

4. Task 3: Microsoft subdomains with basic Linux tools

The HackTarget API returned 51 unique subdomains of microsoft.com. The crt.sh certificate-transparency query failed with a "jq: parse error" because the server returned an error page instead of JSON, so HackTarget was used instead. The free API limit means this is a partial list; microsoft.com has far more subdomains.

Method	Command	Result
Passive API lookup	`curl -s "https://api.hackertarget.com/hostsearch/?q=microsoft.com" | cut -d, -f1 | sort -u > ms_subs.txt`	51 unique subdomains saved in ms_subs.txt
DNS brute force	`for sub in www mail login ...; do host $sub.microsoft.com | grep "has address"; done`	Resolved hosts such as outlook, store and msdn, mostly behind Akamai and Azure addresses
Name servers	`dig microsoft.com NS +short`	ns1-39.azure-dns.com, ns2-39.azure-dns.net, ns3-39.azure-dns.org
Mail servers	`dig microsoft.com MX +short`	microsoft-com.mail.protection.outlook.com (priority 10)
Host lookup	`nslookup www.microsoft.com`	CNAME chain through edgekey.net to e13678.dscb.akamaiedge.net, with IPv4 and IPv6 addresses
Registration data	`whois microsoft.com`	Registry Domain ID 2724960_DOMAIN_COM-VRSN; registrar WHOIS server whois.markmonitor.com

Output files: task3_results.txt (all methods in one file) and ms_subs.txt (subdomain list).

Evidence

Screen recording	Task3/ Screen Recording Task3

5. Task 4: Maltego installation

Maltego 4.13.0 was installed on the Kali VM and launches to its activation screen, where the Maltego ID option is selected.

Commands

```
cd ~/Downloads
chmod +x Maltego*.deb          # give execution permission
sudo dpkg -i Maltego*.deb      # install the package
sudo apt -f install -y         # fix any missing dependencies
maltego                        # launch
```

On first launch Maltego asks how to activate. The "Maltego ID" option (for new users) was chosen, followed by registration, the free Community Edition plan and the terms, after which the main graph window opens.

Evidence

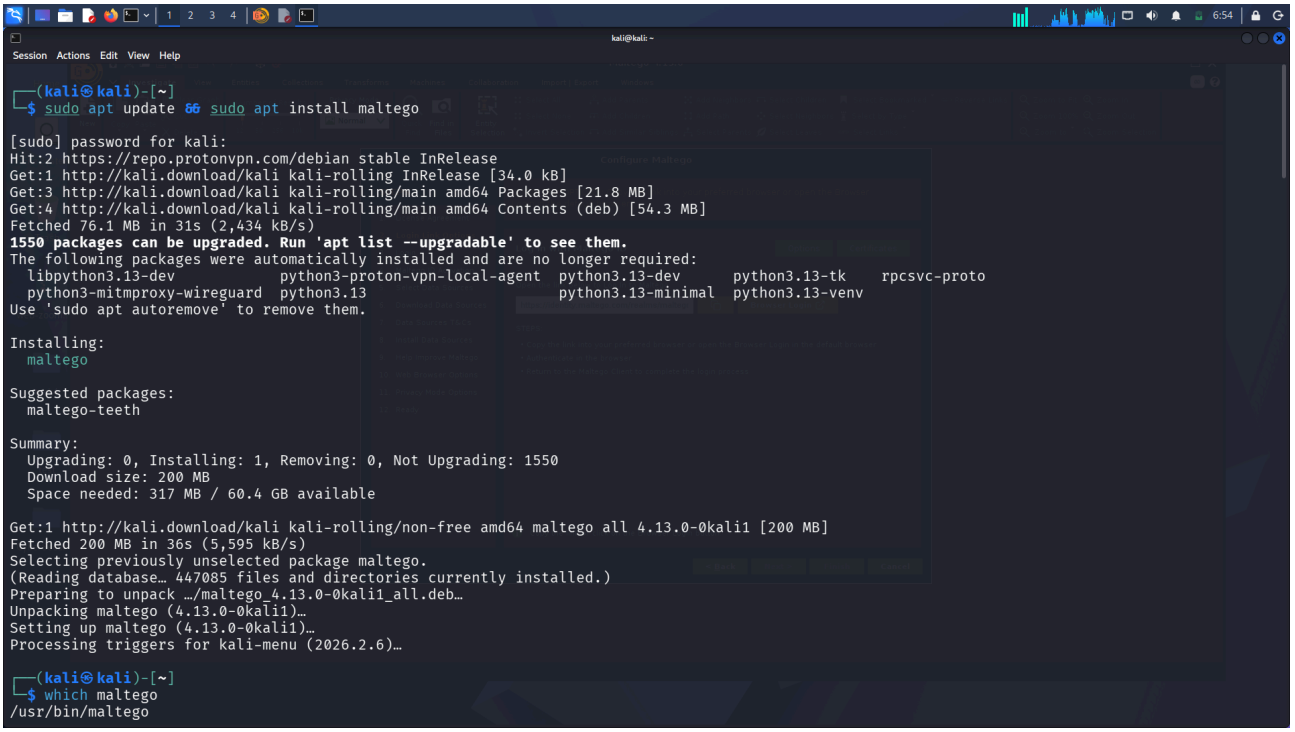


Figure 5.1: execution permission and installation commands

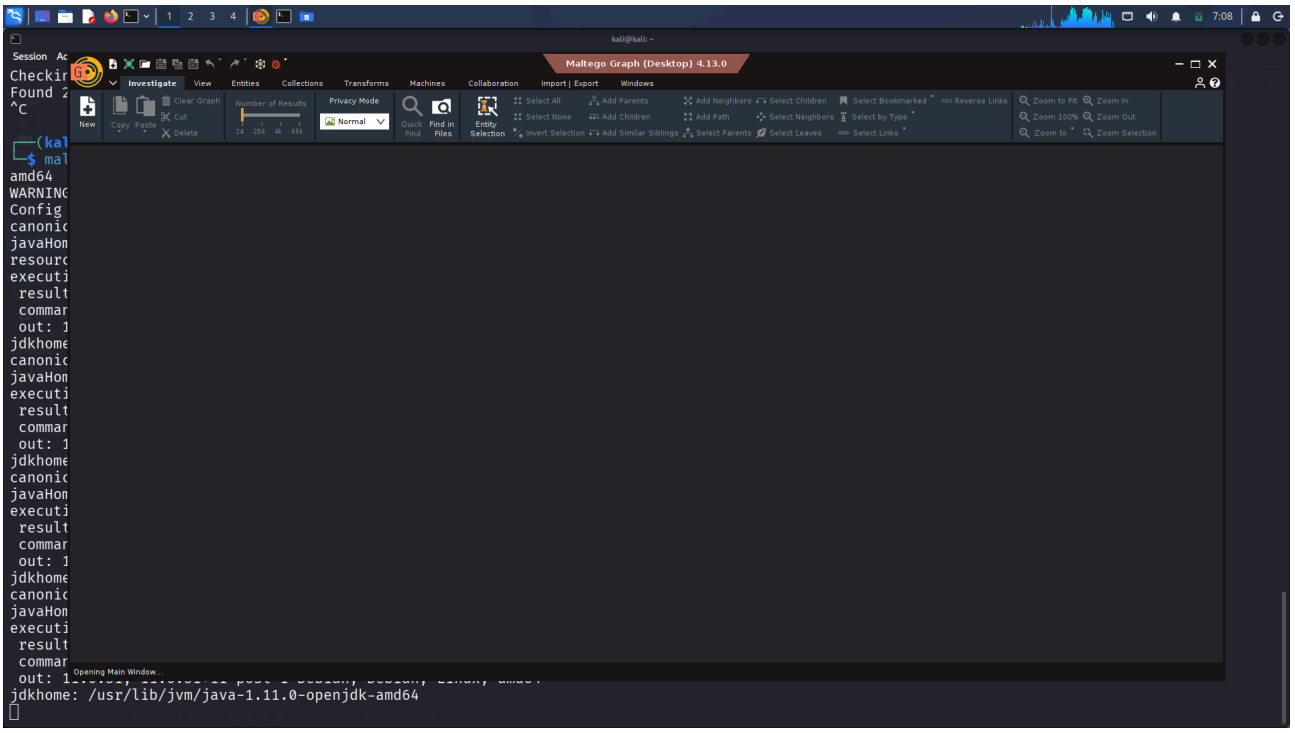


Figure 5.2: Maltego 4.13.0 welcome and activation screen

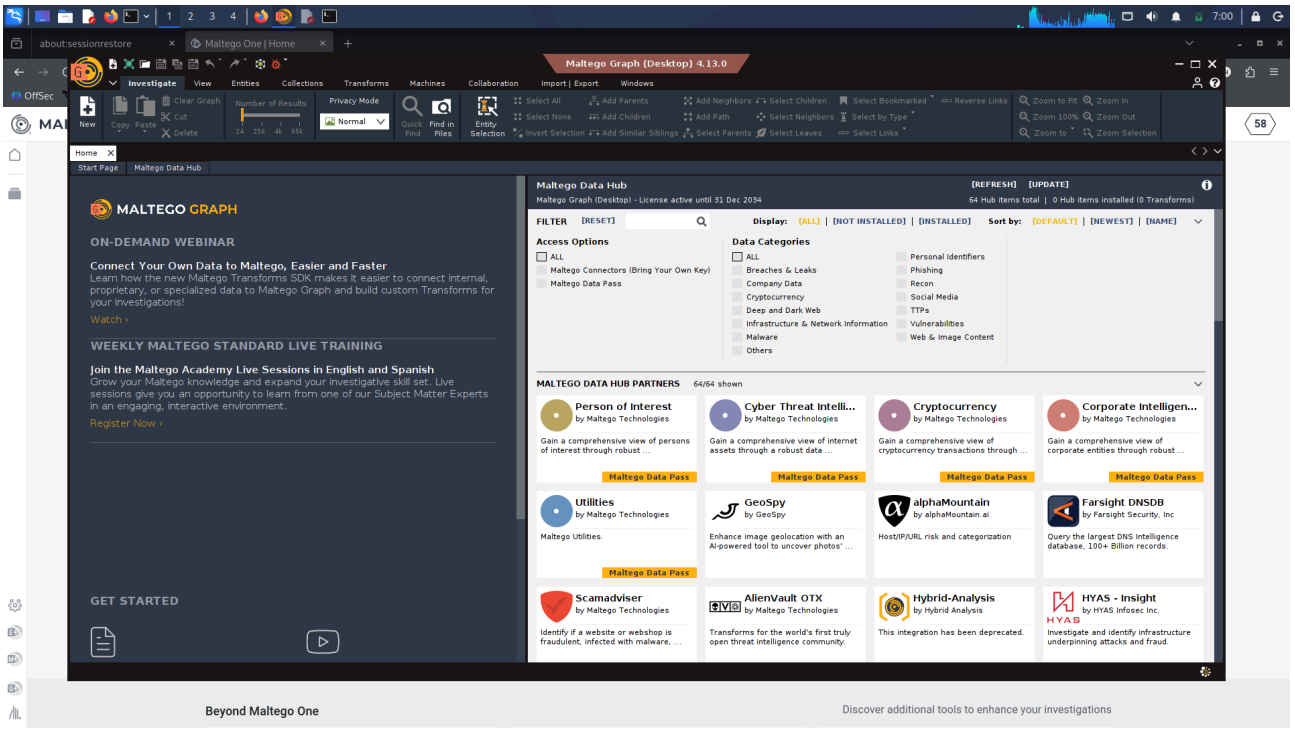


Figure 5.3: Maltego main window after login

6. Task 5: theHarvester with three API keys

theHarvester 4.10.1 was run against microsoft.com with three API keys from the author's own accounts: Shodan, VirusTotal and a GitHub personal access token. Hunter.io and SecurityTrails were skipped because Hunter rejects webmail sign-ups and SecurityTrails has no free API plan. The keys were added to /etc/theHarvester/api-keys.yaml, and the tool read that file at start-up and returned Shodan data for the discovered IP addresses.

Configuration

```
apikeys:
  github:
    key: <GitHub token>
  shodan:
    key: <Shodan key>
  virustotal:
    key: <VirusTotal key>
```

Command

```
theHarvester -d microsoft.com -b shodan,virustotal,github-code -l 100 -f task5_output | tee
task5_results.txt
```

Option	Meaning
`-d`	Target domain
`-b`	Data sources to query
`-l`	Limit on results per source
`-f`	Output file name (creates .json and .xml)

Output files: task5_output.json, task5_output.xml, task5_results.txt

The API keys themselves are not included in this report.

Evidence

Screen recording	Task5/ Screen Recording Task5

7. Conclusion

Public sources alone (certificate and DNS data, search engines and API-backed OSINT tools) exposed subdomains, IP addresses, mail servers and name servers for both targets without touching them directly. Limits came from free tiers: HackTarget returned only 51 subdomains, and crt.sh timed out on a domain as large as microsoft.com.

All work used passive techniques against the domains named in the assignment, with API keys from the author's own accounts. Footprinting like this should only be done against targets one is authorised to test.